

Spatial Pattern of Electric Vehicle Population in State of Washington

DSAN 6750 / PPOL 6805: GIS for Spatial Data Science

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Intro

This project attempts to provide insights into the spatial pattern of EV adoption using approaches from spatial data science.

The Electric Vehicle Population Data provided by Washington State Department of Licensing is the main dataset used in this project. ([“Electric Vehicle Population Data | Data.WA | State of Washington” 2024](#))

Two main sections in this project cover:

Initial Evidence

- Visualization
- Global Moran’s I
- Local Moran’s I

Hypothesis Testing

- Intensity Function
- Relative Intensity Surface
- Monte Carlo Simulation

Set up

packages

Source: [Article Notebook](#)

loading files

Source: [Article Notebook](#)

```
Reading layer `ev1203' from data source
  `/Users/toyuan/24-manuscript/data/ev1203.gpkg' using driver `GPKG'
Simple feature collection with 208002 features and 25 fields
Geometry type: POINT
Dimension:      XY
Bounding box:   xmin: -124.6274 ymin: 45.596 xmax: -117.0455 ymax: 48.99205
Geodetic CRS:   WGS 84
```

```
Reading layer `wa_poly' from data source
  `/Users/toyuan/24-manuscript/data/wa_poly.gpkg' using driver `GPKG'
Simple feature collection with 1 feature and 14 fields
Geometry type: POLYGON
Dimension:      XY
Bounding box:   xmin: -124.849 ymin: 45.54354 xmax: -116.9161 ymax: 49.00244
Geodetic CRS:   NAD83
```

```
Reading layer `urban_shape' from data source
  `/Users/toyuan/24-manuscript/data/urban_shape.gpkg' using driver `GPKG'
Simple feature collection with 56 features and 12 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -124.1802 ymin: 45.78668 xmax: -117.0418 ymax: 49.0021
Geodetic CRS:   WGS 84
```

```
Reading layer `stations' from data source
  `/Users/toyuan/24-manuscript/data/stations.gpkg' using driver `GPKG'
Simple feature collection with 2800 features and 1 field
Geometry type: POINT
Dimension:      XY
Bounding box:   xmin: -124.6629 ymin: 25.91606 xmax: -73.98274 ymax: 48.99526
Geodetic CRS:   WGS 84
```

```
Reading layer `ct1202' from data source
  `/Users/toyuan/24-manuscript/data/ct1202.gpkg' using driver `GPKG'
Simple feature collection with 1784 features and 19 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -124.849 ymin: 45.54354 xmax: -116.9161 ymax: 49.00244
Geodetic CRS:   WGS 84
```

```
Reading layer `ev1203' from data source
  `/Users/toyuan/24-manuscript/data/ev1203.gpkg' using driver `GPKG'
Simple feature collection with 208002 features and 25 fields
Geometry type: POINT
Dimension:      XY
Bounding box:   xmin: -124.6274 ymin: 45.596 xmax: -117.0455 ymax: 48.99205
Geodetic CRS:   WGS 84
```

```
Reading layer `zcta-tl' from data source
  `/Users/toyuan/24-manuscript/data/zcta-tl.gpkg' using driver `GPKG'
Simple feature collection with 605 features and 12 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -124.7365 ymin: 45.54354 xmax: -116.9161 ymax: 49.00244
Geodetic CRS:   WGS 84
```

```
Reading layer `nov-ev' from data source
  `/Users/toyuan/24-manuscript/data/nov-ev.gpkg' using driver `GPKG'
Simple feature collection with 17025 features and 15 fields
Geometry type: POINT
Dimension:      XY
Bounding box:   xmin: -124.6274 ymin: 30.41628 xmax: -71.06739 ymax: 48.99976
Geodetic CRS:   WGS 84
```

Source: [Article Notebook](#)

- Geometries used in this project are from Census Bureau's TIGER/Line Shapefiles
 - State Boundaries([“Index of /Geo/Tiger/TIGER2020PL/STATE/53_WASHINGTON/53” n.d.a](#))
 - Census Tracts([“Index of /Geo/Tiger/TIGER2020PL/STATE/53_WASHINGTON/53” n.d.b](#))
 - Zip Codes([“Index of /Geo/Tiger/TIGER2022/ZCTA520” n.d.](#))
 - Urban Areas([“Index of /Geo/Tiger/TIGER2020/UAC” n.d.](#))

Functions

function to_3857 to_4326 for CRS transform

Source: [Article Notebook](#)

function sf_to_ppp¹

¹to construct marked ppp, extra steps are needed

Source: [Article Notebook](#)

```
function relative_int
```

Source: [Article Notebook](#)

Initial Evidence

This exploratory section approaches EV population in WA as a point pattern, efforts are made to convince that EV adoption is interesting not only as an event but also as a spatial phenomenon. Our approaches includes: landing descriptions of our data, map visualizations and spatial autocorrelation measures.

Visualization

Before presenting our evidence of clustering in the spatial pattern of EVs, we would like to start with some clarification about the “points” and their locations in our data. Each observation in the dataset represents an EV with its unique DOL id. However, the longitude and latitude information corresponds to the centroid of that Zip Code Tabulation Area associated with the recorded address of the EV owner. Thus, 208,002 distinct EVs are matched to only 548 distinct points.

```
ev_counts
```

```
[1] 548
```

Source: [Article Notebook](#)

From the frequency table of counties, we can see that EV population exists across all counties in WA. When plotting EV count per county as a statistic, a skewed distribution gives us the first evidence of the variance in EV adoption counts across regional divisions.

With map visualization of the EV locations and counts in Washington, an impression we might have is that EV adoption as an event does not seem to be constant between locations. Furthermore, when looking at the map, we might suggest detecting clusters in specific locations and, for example, identifying these clusters with the three of the main metropolitan areas that the state overlaps.²([U. C. Bureau n.d.](#))

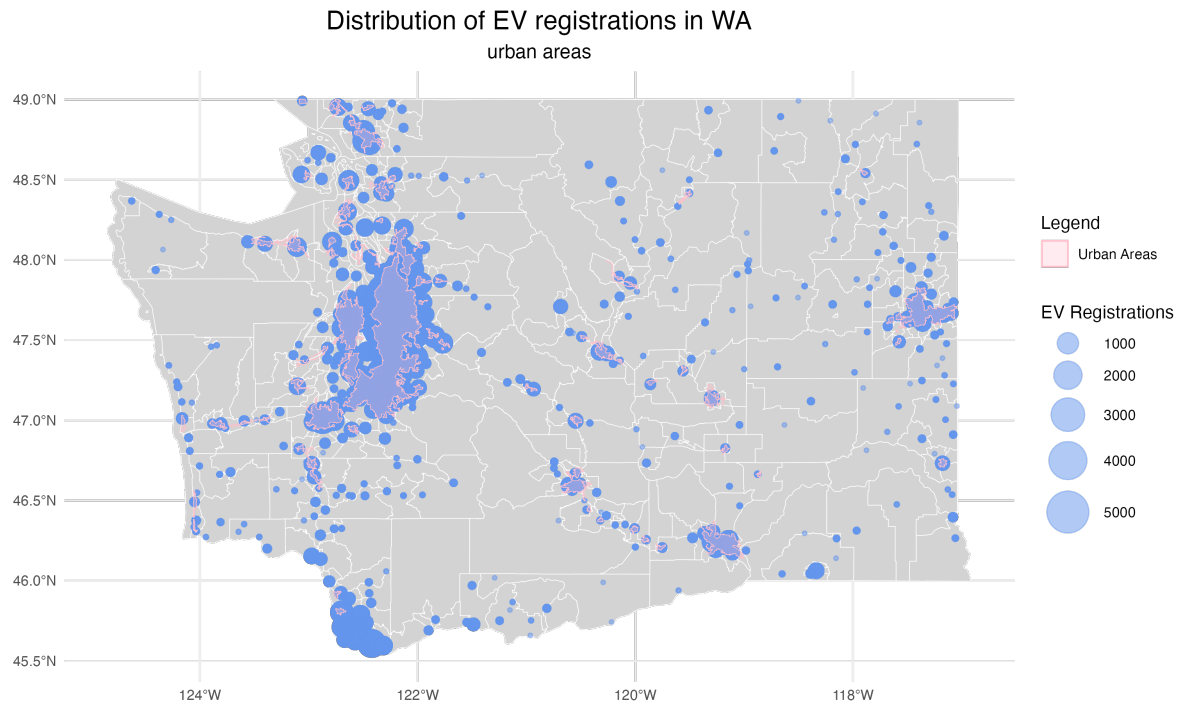
The Seattle-Tacoma-Bellevue Metropolitan area in the northwest; a mid-western cluster corresponding to Spokane-Spokane Valley Metropolitan area; and another approximately in Clark

²Seattle-Tacoma-Bellevue (WA),Portland-Vancouver-Hillsboro (OR-WA), Spokane-Spokane Valley (WA) are the three metropolitan areas listed as the top 100 populous metro areas by the Census Bureau, ranking 15th, 25th and 96th respectively

county, with Vancouver city to take part of the cross-states Portland-Vancouver-Hillsboro Metropolitan area.

```
urb_map <- ggplot() +  
  geom_sf(data = ct_sf, fill = "lightgray", color = "white", size = 0.2) +  
  geom_sf(data = ev_sf, aes(size = count), color = 'cornflowerblue', alpha = 0.5) +  
  geom_sf(data = urban_poly, aes(fill = "Urban Areas"), color = "pink", alpha = 0.3) +  
  scale_size_continuous(name = "EV Registrations", range = c(1, 12)) +  
  scale_fill_manual(name = "Legend", values = c("Urban Areas" = "pink")) +  
  labs(  
    title = "Distribution of EV registrations in WA",  
    subtitle = "urban areas"  
  ) +  
  theme_minimal() +  
  theme(  
    plot.title = element_text(hjust = 0.5, size = 16),  
    plot.subtitle = element_text(hjust = 0.5, size = 12)  
  )  
  
# ggsave("image/manu-urb-ev.png", plot = urb_map)  
urb_map
```

At this stage, we are led to understand the concentration of EV registrations at certain areas, or the attribute similarity of EV counts and their spatial similarity, by suggesting relationships between EV adoption, urban areas and population.



Using the definition provided by US Census Bureau, we can assign urban or rural labels to each registration. We learn from this classification that 180,490 registrations took place in urban areas, leaving 27,512 in rural areas. This spatial unevenness is evident when we impose the polygons of urban areas onto the EV population distribution.

Initial Evidence: Spatial Autocorrelation

Aside from descriptive statistics and mapping, if we seek a more analytical language for communicating our EDA insights, one common approach is to measure spatial autocorrelation for our data.

Global moran's I: Clustering of EV registrations

Firstly, we are interested in measuring the autocorrelation of the whole mapped pattern. The way we determine whether our observed point pattern stands out, is to make comparison with the pattern out of complete spatial randomness. The Global Moran's I statistic offers us a way to measures the autocorrelation of the entire observations.

To estimate the autocorrelation value for our observations using Moran's I formula, we have to specify our statistic of interest, choose a neighbor definition, prepare a neighborhood relation and assign weights to neighbors.

```
function compute_moran
```

Source: [Article Notebook](#)

We tried two ways to use `spdep::moran` for calculating Global Moran's I. The step we take here is to aggregate observed points into polygons, ZIP Code Tabulation Areas which contain 605 polygons, and Census Tracts of 1784 polygons, are two targets of aggregation.³ When constructing the neighborhood relationships, both Rook and Queen definitions of neighbors are tried, and we used the default setting to assign equal weight to all neighbors. Both attempts are to result in a positive autocorrelation of our observations, pointing us in the direction of non-random spatial clustering.

However, the positivity of this autocorrelation value is not that fruitful on its own. For us to reject CSR or to make sense of the significance of our observed Moran's I statistic, we further did a permutation test using `spdep::moran.mc()` and obtained a pseudo p-value of 0.001. Since this is the most extreme value we can get in the case of 999 Monte Carlo Simulations, the likelihood that we reject CSR while CSR is true is a low one. Given a positive and significant global moran's I, we can say that our observe point pattern differs significantly from a random point pattern in the direction of clustering. ([R Consortium 2022](#))

³The reason that I did not simply use ZCTAs is although the codebook describes the location variable as the center of postal code areas, but the provided longitude and latitude did not align well with the centroids of ZCTAs from tiger/line shapefile, with cases that more than one points fell within a single ZCTA

Moran's I using ZCTAs as bounding polygons and Queen Definition of Neighbors

Source: [Article Notebook](#)

Source: [Article Notebook](#)

```
[1] 0.6457255
```

Source: [Article Notebook](#)

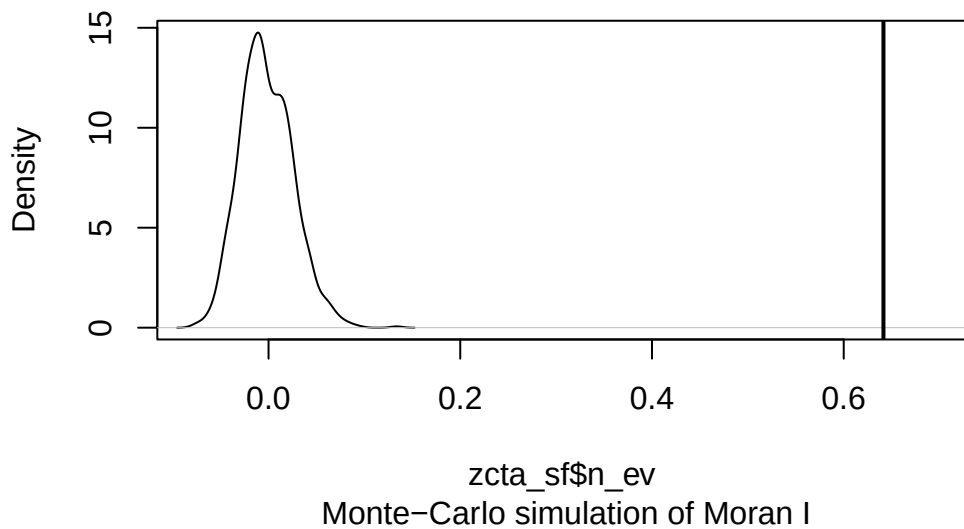
Monte-Carlo simulation of Moran I

```
data:  zcta_sf$n_ev
weights: zcta_listw
number of simulations + 1: 1000

statistic = 0.64146, observed rank = 1000, p-value = 0.001
alternative hypothesis: greater
```

Source: [Article Notebook](#)

Density plot of permutation outcomes



Source: [Article Notebook](#)

Moran's I using Census Tracts as bounding polygons and Rook definition of neighbors

```
[1] 208002
```

Source: [Article Notebook](#)

Source: [Article Notebook](#)

Moran's I value

```
[1] 0.5136527
```

Source: [Article Notebook](#)

visualize neighborhood relation

Source: [Article Notebook](#)

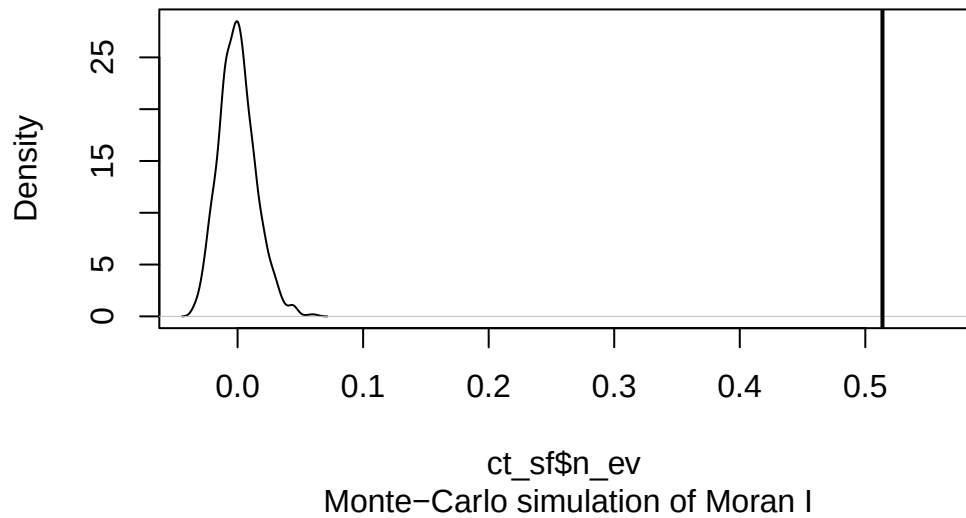
Monte-Carlo simulation of Moran I

```
data:  ct_sf$n_ev
weights: ct_listw
number of simulations + 1: 1000

statistic = 0.51365, observed rank = 1000, p-value = 0.001
alternative hypothesis: greater
```

Source: [Article Notebook](#)

Density plot of permutation outcomes



Source: [Article Notebook](#)

Local Moran's I: detecting clusters

With evidence of clustering globally, we also want information about where we can find the clusters. In this case, `spdep::localmoran` can help use with a Moran's I statistic for each census tract. ([GeoDa Software 2021](#))

```
[1] "localmoran" "matrix"      "array"
```

```
[1] 1784      5
```

Source: [Article Notebook](#)

Source: [Article Notebook](#)

```
[1] 1784     26
```

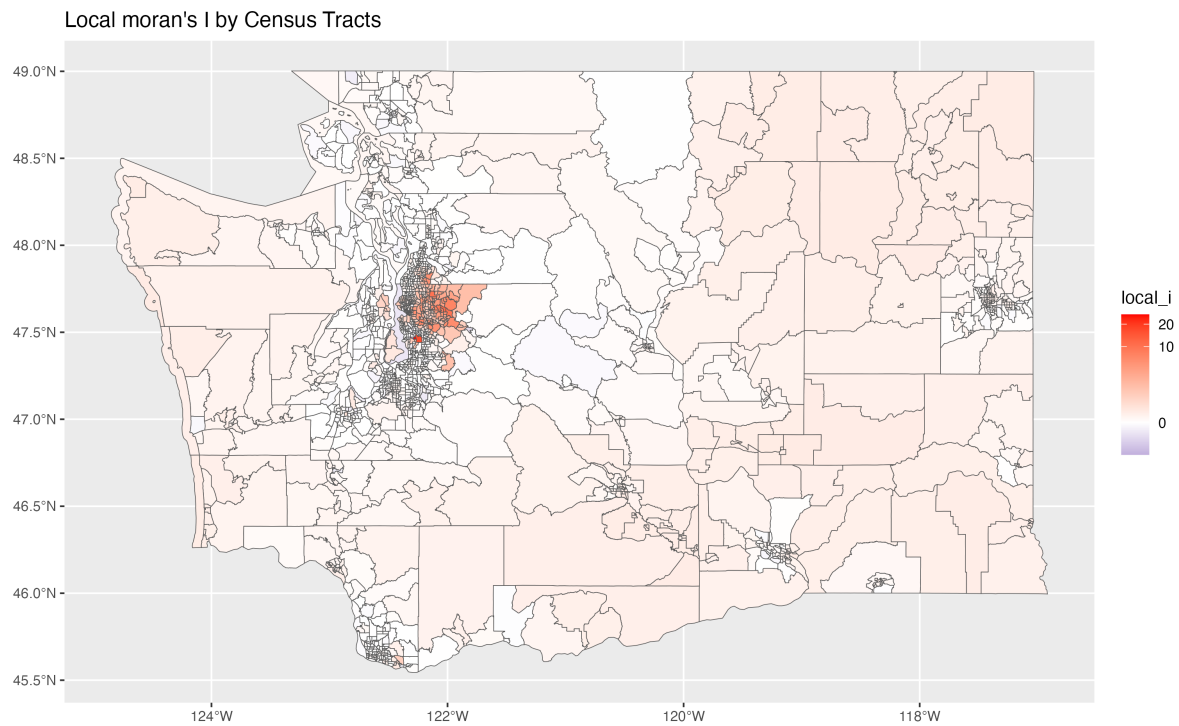
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-2.28334	0.03044	0.19644	0.51365	0.46033	26.79675

Source: [Article Notebook](#)

```

ev_ct_local <- ct_sf7 |> ggplot() +
  geom_sf(aes(fill= as.numeric(local_i))) +
  labs(
    title = "Local moran's I by Census Tracts",
    fill = "local_i"
  ) +
  scale_fill_gradient2(low="darkblue", high="red",
    transform = "pseudo_log")
# +
  scale_fill_viridis_c(na.value = "gray")
# ggsave('image/ct_localmoran.png', ev_ct_local)
ev_ct_local

```

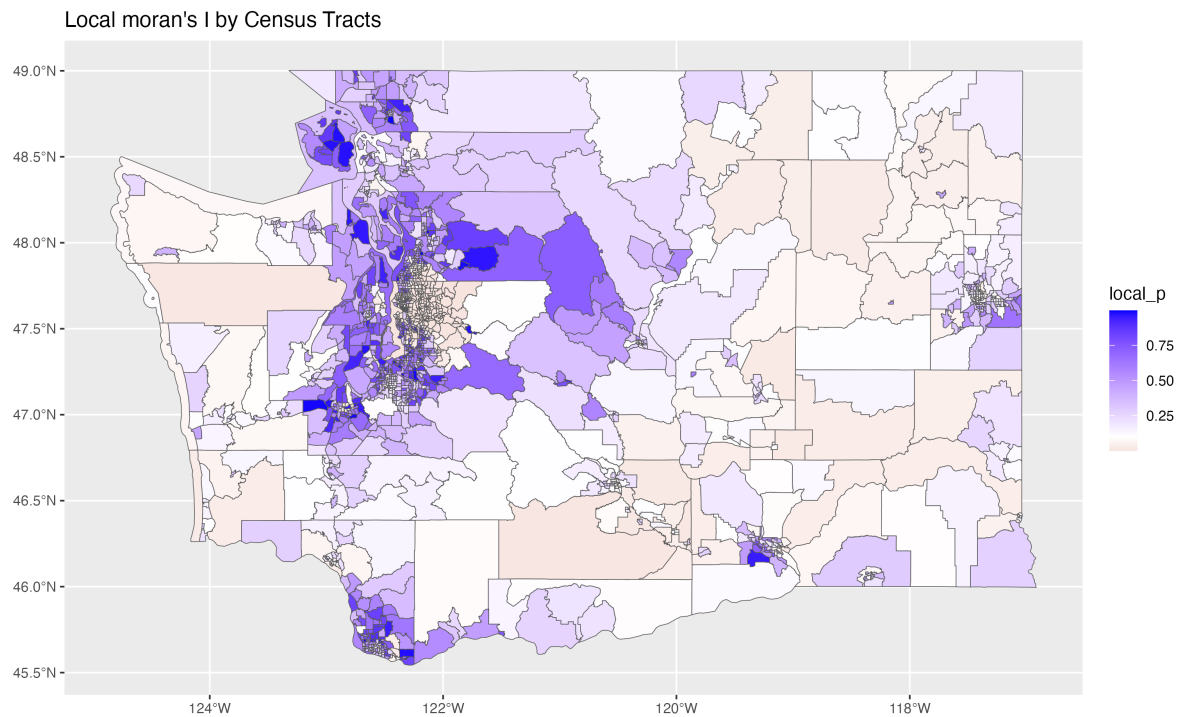


Local Moran's I significance map

```
ev_ct_local_p <- ct_sf7 |> ggplot() +
  geom_sf(aes(fill= as.numeric(local_p))) +
  labs(
    title = "Local moran's I by Census Tracts",
    fill = "local_p"
  ) +
  scale_fill_gradient2(low="darkred", high="blue", midpoint=0.1)
```



```
# +
  scale_fill_viridis_c(na.value = "gray")
# ggsave('image/ct_localmoran_p.png', ev_ct_local_p)
ev_ct_local_p
```



local moran permutation test

```
[1] "localmoran" "matrix"      "array"
```

[1] 1784 9

Source: [Article Notebook](#)

	Ii	E.Ii	Var.Ii	Z.Ii	Pr(z != E(Ii))	
1	-0.19651842	-0.0031906570	6.079843e-02	-0.7840577	0.4330062	
2	0.00244076	0.0005481923	6.916542e-05	0.2275657	0.8199839	
	Pr(z != E(Ii))	Sim	Pr(folded)	Sim	Skewness	Kurtosis
1		0.406		0.203	2.272828	12.18684
2		0.934		0.467	-2.498880	18.68960

Source: [Article Notebook](#)

From the conditional permutation test, we can plot local significance map and specify different types of associations. The table below is a count table for a Moran scatter plot, we can see that among 521 census tracts with significant pseudo p-values 306 census tracts exhibit Low-Low associations, 205 census tracts for High-High associations.([Bivand et al. 2017](#))

Low-Low	High-Low	Low-High	High-High	<NA>
306	8	15	205	1250

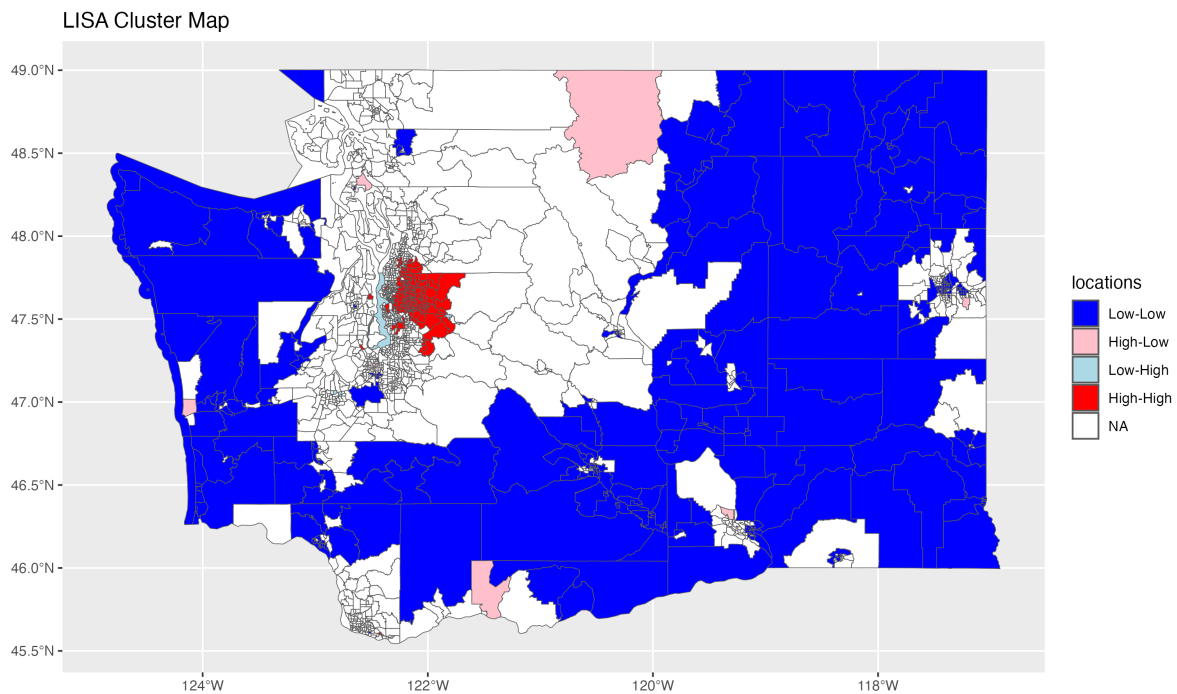
Source: [Article Notebook](#)

Source: [Article Notebook](#)

LISA Cluster Map

```
local_perm_map <- ct_sf7 |> ggplot() +  
  geom_sf(aes(fill = scatter)) +  
  labs(  
    title = "LISA Cluster Map",  
    fill = "locations"  
  ) +  
  scale_fill_manual(  
    values = c(  
      'High-High' = 'red',  
      'High-Low' = 'pink',  
      'Low-High' = 'lightblue',  
      'Low-Low' = 'blue'  
    ),  
    na.value = 'white'
```

```
)
local_perm_map
# ggsave('image/manu-lisa-cluster.png', local_perm_map)
```



Local Moran's I, pseudo p-values from the conditional permutation test, and identification of the association types for the significant census tracts, take us to a more in-depth understanding of our EV point pattern.

Insight into the hotspots and coldspots of EV adoption, corresponding to High-High census

tracts and Low-Low census tracts is especially helpful because we are inclined to see the hotspots but ignore the coldspots. The LISA Cluster Map reveals clusters centered census tracts with low EV population themselves are also surrounded by neighbors with low EV population. This finding is hardly possible to get if we only have the positive cases, or already EV owning households to visualize. An important aspect of interpreting positive spatial autocorrelation is demonstrated here: we need to think of the bigger picture of clustering as the contribution of attribute (in our case EV count) similarity and locational similarity, thus, we won't ignore clusters that have similar low EV counts also have a role to play in this picture. ([Anselin 2020](#))

Hypothesis Testing

This section attempts to deal with another aspect we need to account for when acquiring a positive and significant global autocorrelation value for our point pattern of interest. In our case, we want to evaluate how the spatial inhomogeneity of the Washington state can explain for our initial evidence of clustering.

We want to understand the first-order properties of EV adoption pattern with the awareness of the underlying landscape of the state that can have an effect on the adoption potential of different locations. We can take a look at our EV population distribution together with the human population distribution in WA. They appeared to follow very similar distribution, and we do recognize that the core of the definition of an urban area is the spatial distribution of population.

Thus, it will be helpful to set aside other characteristics we associate with the urban label at the time, examine the relationship between the two populations in WA. We expect a better understanding of this relationship can guide us to explore to which extent that the EV counts variations across locations can be explained by the population count variations, and how population distribution as an explanation can have different degrees of effectiveness from one region to another.

```
distribution_map <- ggplot() +  
  geom_sf(data = ct_sf, fill = "lightgray", color = "white", size = 0.2) +  
  geom_sf(  
    data = ev_sf,  
    aes(size = count,  
        color = cb_palette[5]),  
    alpha = 0.6) +  
  scale_size_continuous(name = "EV Registrations", range = c(0.1, 15)) +  
  labs(  
    title = "Distribution of ZEVs in WA",  
    subtitle = "Cumulative data to October 31, 2024",  
    x = "longitude",
```

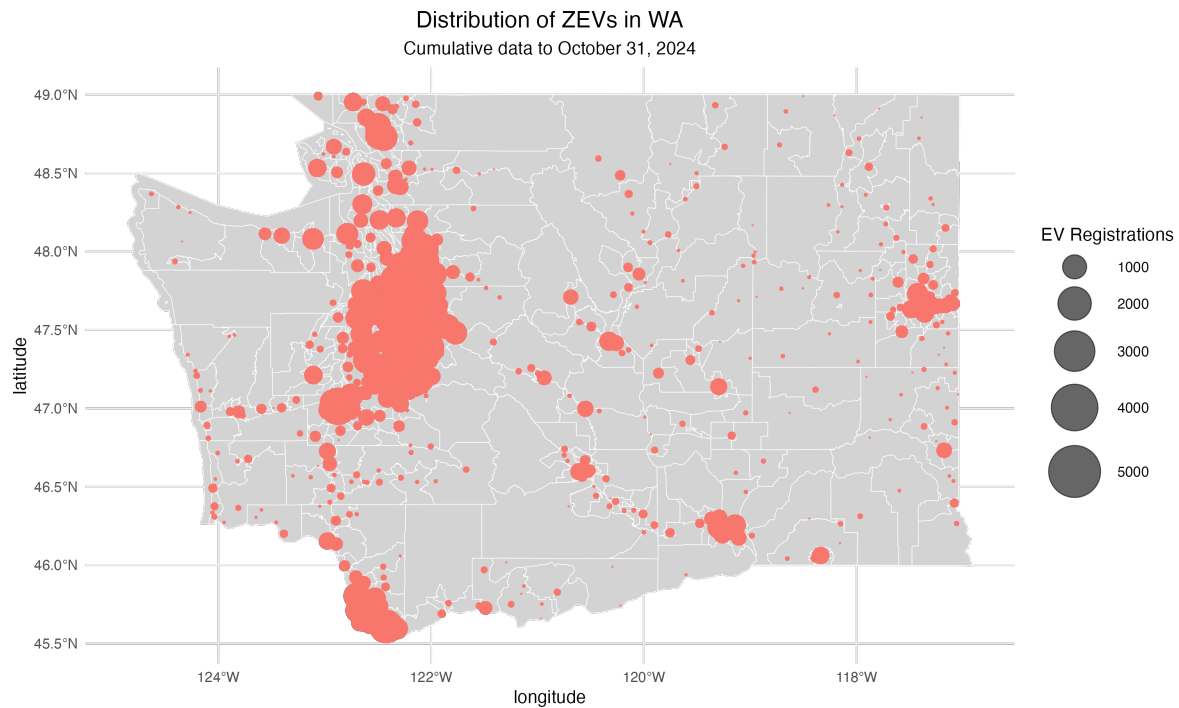
```

    y = "latitude"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 11)
  ) +
  guides(color = "none")

ggsave("image/manu-ev-distribution.png", plot = distribution_map)

distribution_map

```



NASA GPW data provides us a glimpse of disaggregated human population data, modeling a human population count for each 1 km^2 grid. We can take it conveniently as a way to make sense of what population point pattern can look like.⁴([International Earth Science Information Network \(CIESIN\) - Columbia University 2018](#))

`gpw_raster`

⁴Center for International Earth Science Information Network - CIESIN - Columbia University (2018). Gridded Population of the World, Version 4 (GPWv4): Population Count Adjusted to Match 2015 Revision of UN WPP Country Totals, Revision 11. Palisades, NY: NASA Socioeconomic Data and Applications Center (SEDAC). doi: 10.7927/H4PN93PB. Accessed December 4, 2024.

```

class      : RasterLayer
dimensions : 416, 955, 397280  (nrow, ncol, ncell)
resolution : 0.008333333, 0.008333333  (x, y)
extent     : -124.8667, -116.9083, 45.54167, 49.00833  (xmin, xmax, ymin, ymax)
crs       : +proj=longlat +datum=WGS84 +no_defs
source    : gpw-popcount.tif
names     : gpw.popcount

```

Source: [Article Notebook](#)

Source: [Article Notebook](#)

```
[1] 304514      2
```

Source: [Article Notebook](#)

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000	0.000	0.144	24.585	1.504	7425.167

Source: [Article Notebook](#)

```

gpw_map <- popcount_sf |> ggplot(aes(fill = pop)) +
  geom_sf(color = scales::alpha("white",0)) +
  scale_fill_viridis_c(trans = 'pseudo_log') +
  theme_minimal() +
  labs(
    title = "Population Counts in WA",
    subtitle = 'GPW V.4 2020-07-01',
    fill = "Population Count"
  ) +
  theme(
    plot.title = element_text(hjust=0.5),
    plot.subtitle = element_text(hjust=0.5)
  )

ggsave('image/manu-gpw-raster.png', plot = gpw_map)

gpw_map

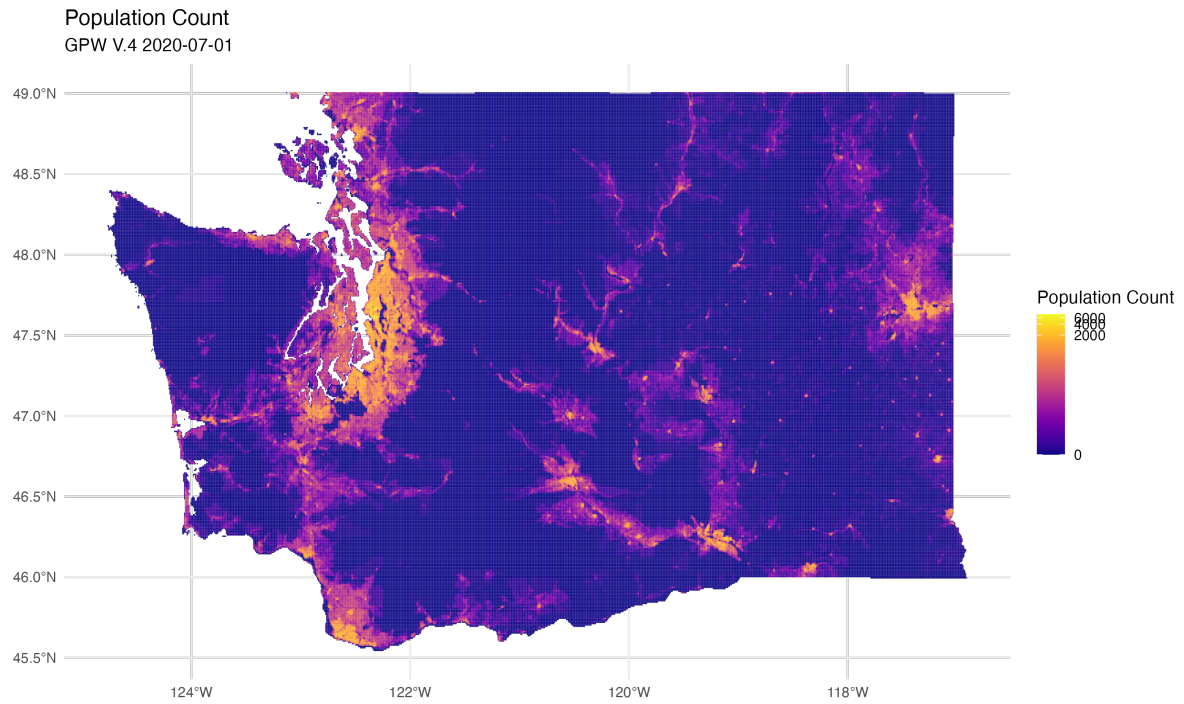
```

```

un_map <- popcount_sf |> ggplot(aes(fill = pop)) +
  geom_sf(color = NA) +
  scale_fill_viridis(option = 'C', trans = 'pseudo_log') +
  theme_minimal() +
  labs(
    title = 'Population Count',
    subtitle = 'GPW V.4 2020-07-01',
    fill = 'Population Count'
  )

ggsave('image/manu-gpw-0701.png', plot = un_map)
un_map

```

Our null hypothesis is: the observed EV point pattern can be directly explained by the population distribution of WA. In other words, people in WA have equal chance to adopt EV regardless of where they live, rural or urban and also regardless of whether their neighbors own EVs or not.

First order properties of point process

Intensity Function

One important limitation of this analysis that needs to be addressed is that we use the WA state boundaries as our observation window. The drawback of this choice is quite obvious: among the three major metropolitan areas in WA, two are situated at the borders of the state – one with Oregon and the other with Idaho. This can lead to underestimations, as they can have less neighbors as the result of this bounding box.

Intensity function we are interested here is the count of event: count of EVs and count of people in an area

Constructing the `ppp` object for estimating EV population intensity is a more straightforward one, we use the 548 distinct points and marked them with EV registration counts. When it comes to finding an appropriate way to estimate human population intensity, since ZCTAs are not a typical statistical division of US Census Bureau, thus, one way to approach this problem is to use the data of population counts by census tract and use the centroids of the census tracts to construct `ppp` object. Another way we explored is to take the 605 centroids of ZCTAs and request points estimate of population counts also from NASA APPEEARs. ([International Earth Science Information Network \(CIESIN\) - Columbia University 2018](#))

With these `ppps` at hand `spatstat::density()` can handle these marked `ppp` and estimate us the intensity function accounting the marks as weights.

zev_ppp

Source: [Article Notebook](#)

Source: [Article Notebook](#)

Source: [Article Notebook](#)

zev_int



Source: [Article Notebook](#)

```
[1] 70971.82
```

```
[1] 1.490787e-07
```

Source: [Article Notebook](#)

pop_int

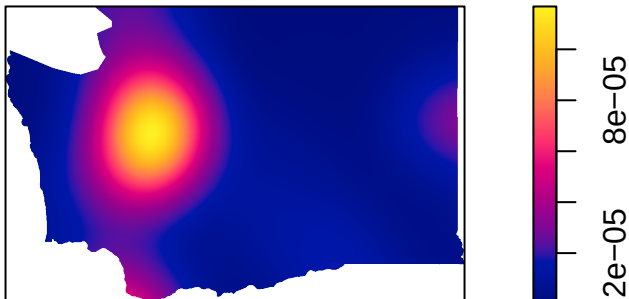
- The data used for population counts by Census Tract([U. S. C. Bureau, n.d.](#))

Source: [Article Notebook](#)

```
[1] 7705281
```

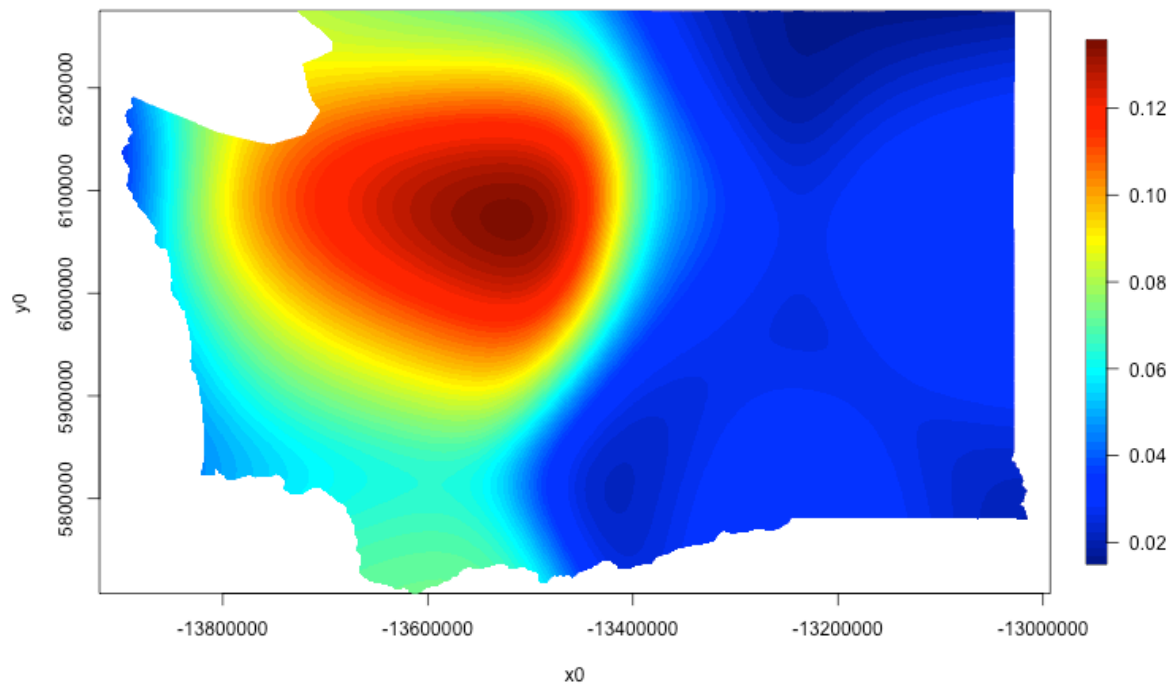
Source: [Article Notebook](#)

pop_int



Source: [Article Notebook](#)

```
# png('image/manu-ct-ratio.png', width = 750, height = 500)
relative_int(zev_ppp, pop_ppp)
# dev.off()
```



With these ppps at hand `spatstat::density()` can handle these marked ppp and estimate us the intensity function accounting the marks as weights.

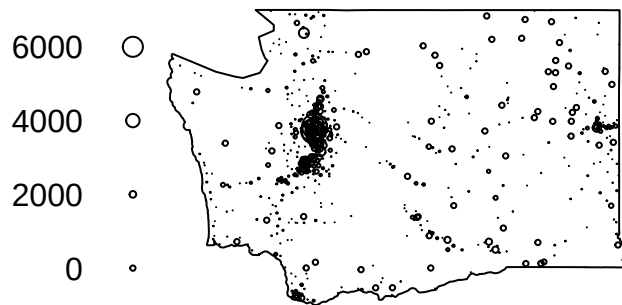
gpw_ppp

Source: [Article Notebook](#)

Source: [Article Notebook](#)

Source: [Article Notebook](#)

gpw_ppp

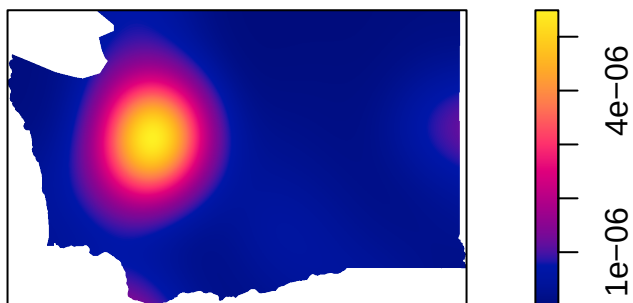


Source: [Article Notebook](#)

```
[1] 308385.8
```

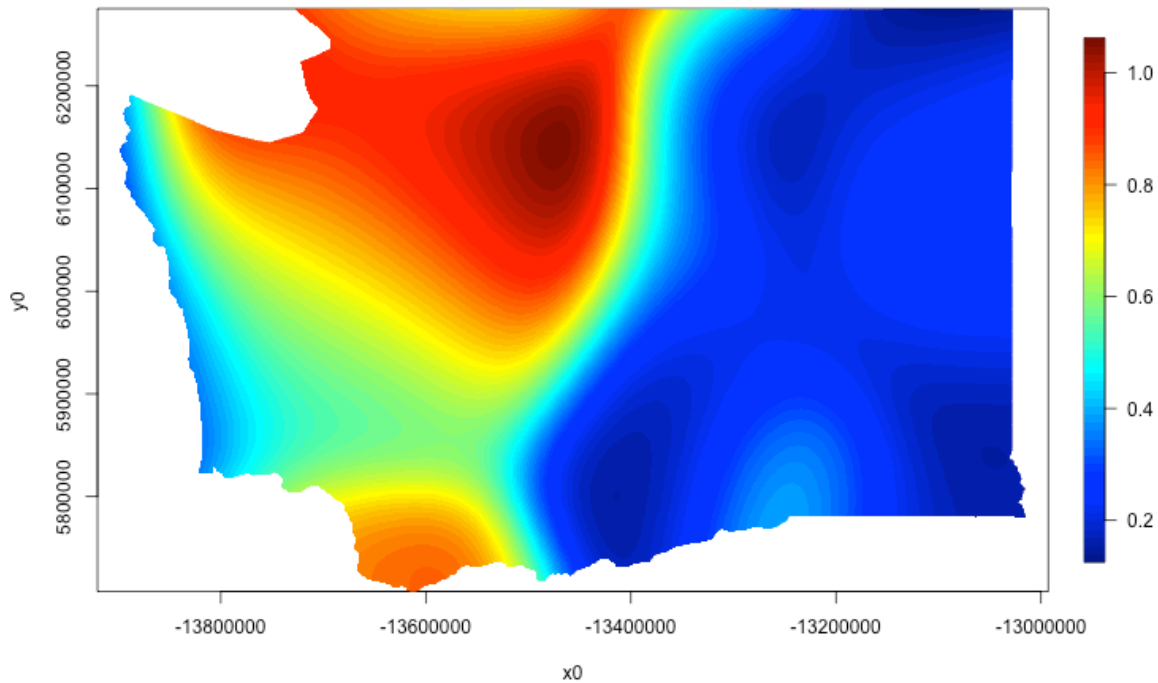
Source: [Article Notebook](#)

gpw_int



Source: [Article Notebook](#)

```
# png('image/manu-zpts-ratio.png', width = 750, height = 500)
relative_int(zev_ppp, gpw_ppp)
# dev.off()
```



Relative Intensity Surface

We use the optimal bandwidths given by spatstat and make adjustments to calculate intensity ratios between our case point pattern to the base rate point pattern. By eliminating the effect of the human population intensity of WA, the relative intensity surface plots still exhibit uneven ratio from location to location suggesting us that point pattern of EV population is a special one.⁵(Moraga n.d.)

Monte Carlo Simulation

We deploy quadrat count to understand how intensity varies from dividing the state in to equal regions of “Low”, “Medium” and “High” population. Two intensity functions constructed by Census tract level population count and GPW population count are used as the null point process in the Monte Carlo simulation.

⁵the interpretation of the ratio values have things to do with the different number of points between ppp objects, but the color variances is interpretable

divide regions for quadrat count
pop_areas



Source: [Article Notebook](#)

computational approach: 999 simulations

```
set.seed(6805)
gen_sims_ppp <- function(num_sims = 999) {
  ev_sims <- spatstat.random::rpoint(
    n = nrow(ev_sf),
    f = pop_int,
    nsim = num_sims
  )
  return(ev_sims)
}
n_sims <- 999
ev_sims_list <- gen_sims_ppp()
```

Source: [Article Notebook](#)

calculate test statistic from observations

Low	Medium	High
1644	9001	197357

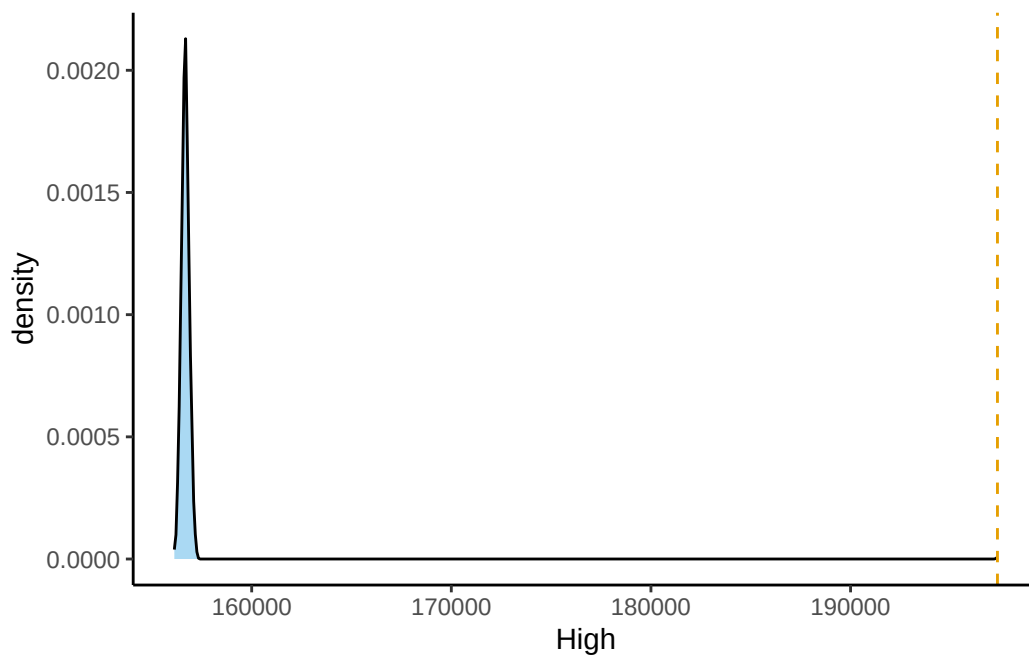
Source: [Article Notebook](#)

calculate test statistic from simulations

```
sims_region_counts <- lapply(  
  X = ev_sims_list,  
  FUN = compute_quadrat_counts  
)  
sim_counts_df <- as_tibble(sims_region_counts) |> t() |> as_tibble()  
colnames(sim_counts_df) <- region_labels  
sim_counts_df |> head(4)
```

```
bind_df <- bind_rows(sim_counts_df, obs_counts)  
bind_df |> dim()  
bind_df |> write_csv('data/manu-simulations.csv')
```

Source: [Article Notebook](#)



Source: [Article Notebook](#)

```
mc_sim <- function(im, fpath1, fpath2, num_sims = 999){  
  n_regions <- 3  
  region_labels <- c("Low", "Medium", "High")  
  int_vals <- im  
  int_quant <- quantile(int_vals, probs = ((0:n_regions) / n_regions), na.rm = TRUE)  
  int_cut <- cut(int_vals, breaks = int_quant, labels = region_labels)
```



```

int_areas <- tess(image = int_cut)
plot(int_areas)

set.seed(6805)
gen_sims_ppp <- function(num_sims) {
  spatstat.random::rpoint(
    n = nrow(ev_sf),
    f = im,
    nsim = num_sims)
}

sims_list <- gen_sims_ppp(num_sims)

compute_quadrat_counts <- function(sim_ppp) {
  counts <- quadratcount(sim_ppp, tess = int_areas) |> as.vector()
  names(counts) <- region_labels
  return(counts)
}

ev_ppp <- sf_to_ppp(ev_sf)
obs <- compute_quadrat_counts(ev_ppp)

sims_region_counts <- lapply(
  X = ev_sims_list,
  FUN = compute_quadrat_counts
)
sim_counts_df <- as_tibble(sims_region_counts) |> t() |> as_tibble()
colnames(sim_counts_df) <- region_labels

bind_df <- bind_rows(sim_counts_df, obs_counts)
bind_df |> write_csv(fpath1)

sims_plot <- bind_df |>
  ggplot(aes(x = High)) +
  geom_density(fill = cb_palette[2], alpha = 0.5) +
  geom_vline(
    xintercept = obs['High'],
    linetype = "dashed",
    color = cb_palette[1]
  ) +
  theme_classic()

```

```

ggsave(fpath2, plot = sims_plot, width = 14, height = 6)

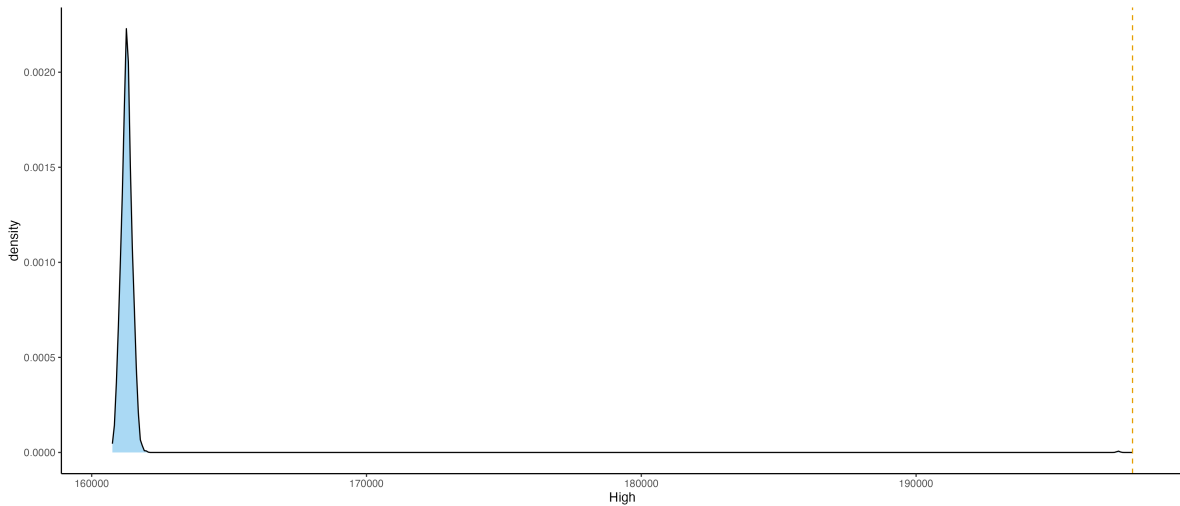
return(sims_plot)
}

```

```

mc_sim(gpw_int, 'data/gpw-sims.csv', 'image/gpw-distribution.png')

```



The computational approach gives us an entire distribution of our statistics of interest, under the simulations of distributing 208002 EV registrations by human population point process. Given that the observed value lies on the far right tail of the distribution, the likelihood for us to observe 197,900 EV registrations in high population regions is a very low one. Therefore, we can believe that the null hypothesis is not likely to stand.

The significance of our observed statistics in high population areas is more like a general picture that justifies our attentions to the spatial phenomenon of EVs, however, the findings from LISA Cluster Map from the relative intensity ratio do ask us to pay attention to local differences and how these local differences can contribute to the characteristics of the whole point pattern.

Discussion

With the evidence of clustering and after extracting the effect of population intensity, we have evidence to describe EV population point pattern in WA as a whole spatially clustering, and to think about the implications of this overall unevenness. At the same time, the local perspective offers insights into the relationship between attributes and locations in significant clusters.

Knowing the adoption pattern and characteristics of those who already owns an EV might be a way of analyzing the issue of electric vehicle, if the aim of the issue is to anticipate a wider adoption and to eliminate the inequality, getting the glimpse of Low-Low clusters in many ways can be even more insightful. Another way to think of this issue is that of EV owning households today assumingly already have a positive view on EV and they made the decision of buying an EV rationally after evaluating their own living circumstances. Thus, to promote a more positive and decent view on electric vehicles and to prepare a more friendly environment that can support EV purchases, it is evident that more efforts should be made for the Low-Low clusters.

Conclusion

One perspective confirmed by this project is that, it is actually not that hard to discover EV hotspots, we can find them generally in the urban areas or metro areas, though they may have different degrees of concentration. For example, the heart of Seattle Metro area, census tracts in King County out-performed both the CRS distribution and population distribution. In the case of Clark county and Spokane city, the results are less significant, however, may be due to the effect of the observation window. On the other hand, detecting coldspots requires many more steps, and we tend to underestimate the contribution of local Low-Low clusters to the global clusterness. But coldspots are indeed part of—and might be the essence of—the inequality behind the uneven spatial pattern of EV adoption.

Most of the analysis in this project is at the exploratory stage. As the last section of the project, we are to think about where the EDA results can take us.

Two of the main reason that people usually give for not considering an EV are: EVs are not affordable, and that EVs are not reliable in terms of the lack of access to charging equipment. (Kikuchi 2024)

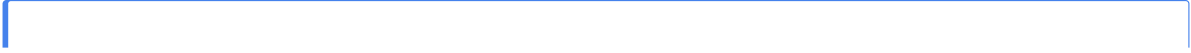
Income and charging station accessibility, at a glance are both following spatial distributions similar to EV population and urban-rural divisions. Research on EVs has devoted much efforts carrying out the message that, with possible rebates to make the upfront cost of purchasing an EV less burdensome, EVs can, in the long term, save commuters more money than petro and diesel cars. As for the range anxiety, which is more expressed by potential EV owners than by actual EV owners, it is explained by that most recent EV owners charge their vehicles at home on daily bases. This implies that they have private parking spaces, can afford to install home charging equipment, and the electric range of their EVs is just sufficient for their daily commute distances. (“Alternative Fuels Data Center: Charging Electric Vehicles at Home” n.d.)

But these facts of today are not there to convince us that affordability and public charging infrastructure don’t really matter much in the decision of buying an EV. Instead, they can lead us to see that the diffusion of EVs at a stage where barriers between owners and potential owners are not only material but also in knowledge.

Looking forward to the 2030 ZEV mandate, all newly sold vehicles electric, one solution is to make the most of EV charging station. An analysis by Pew Research Center offers an interesting way to think about public charging stations.([Shah 2024](#)) Or, people need to see charging stations for they to develop confidence in EV before they really use them or even before people actually own an EV.

Addressing the unevenness of adoption and the importance of the visibility of EV charging station in the EV diffusion process, can be seen in how Induction Reduction Act incentives target EV charging stations in rural areas, and in reseach exploring models to determine the optimal locations for EV charging station installations. ([Needell et al. 2016](#))

The following note is an attempt to take the previous approaches, but this time analyzing the distribution of EV charging station. Some early findings from the relative intensity ratio of charging stations on EV registrations are that the charging station distribution is somewhat more even compared to the EV registrations.



For the interpretation of the interesting census tracts we find in the LISA cluster map, there is much to anticipate if domain knowledge of the state and locals is introduced.

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